EDEXCEL INTERNATIONAL GCSE MATH (4MA1) • UNIT 8.01

C Histograms, Cumulative Frequency & Conditional Probability

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

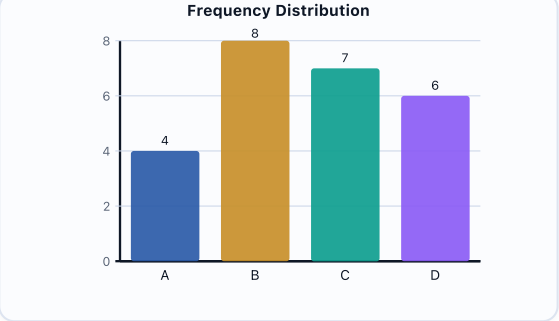
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 D	Q2 A	Q3 B	Q4 C	Q5 D
Q6 A	Q7 B	Q8 C	Q9 C	Q10 B
Q11 A	Q12 B	Q13 A	Q14 B	Q15 B
Q16 A	Q17 D	Q18 D	Q19 C	Q20 B
Q21 A	Q22 B	Q23 D	Q24 B	Q25 C

Histograms, Cumulative Frequency & Conditional Probability Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 20x + 118$ in the domain of Histograms, Cumulative Frequency & Conditional Probability. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(12)$.



A 50

B 22

C 38

D 40 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Histograms, Cumulative Frequency & Conditional Probability):

1. Complete the square: $x^2 - 20x + 118 = (x - 10)^2 - 100 + 118 = (x - 10)^2 + 18$, giving minimum value $k = 18$. **[M1]**

2. Evaluate at $x = 12$: $f_1(12) = (2)^2 + 18 = 22$. **[M1]**

3. Sum the two quantities: $k + f_1(12) = 18 + 22 = 40$. **[A1]**

- **Final Verified Answer:** 40.

2 MCQ

A mathematical model in Histograms, Cumulative Frequency & Conditional Probability is defined by the cubic function $y_2(x) = 3x^3 - 10x^2 + 15x - 15$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

- **A** 36 ✓
- **B** 96
- **C** 21
- **D** 22

WORKED SOLUTION & VERIFICATION

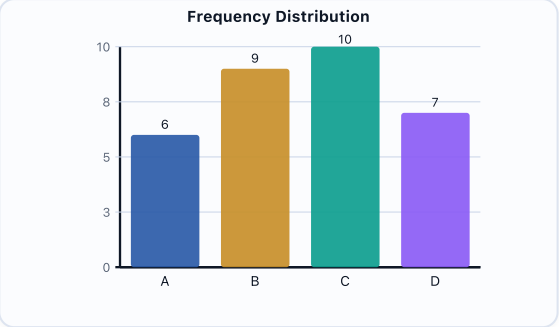
Mark Scheme & Examiner Solution (Histograms, Cumulative Frequency & Conditional Probability):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 9x^2 - 20x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 9(9) - 20(3) + 15 = 36$. **[M1][A1]**

- **Final Verified Answer:** 36.

3 MCQ

In Histograms, Cumulative Frequency & Conditional Probability, two displacement vectors are given by $\mathbf{u}_3 = \langle 10, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.



- **A** 34
- **B** 198 ✓
- **C** 164
- **D** 310

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Histograms, Cumulative Frequency & Conditional Probability):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (10)(9) + (-8)(7) = 90 + (-56) = 34$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 10^2 + (-8)^2 = 164$. **[M1]**
3. Combine: $34 + 164 = 198$. **[A1]**

- **Final Verified Answer:** 198.

4 MCQ

The linear growth profile shown in the diagram for Histograms, Cumulative Frequency & Conditional Probability follows an arithmetic progression with first term $a_1 = 11$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 95

B 885

C 795 ✓

D 1425

WORKED SOLUTION & VERIFICATION

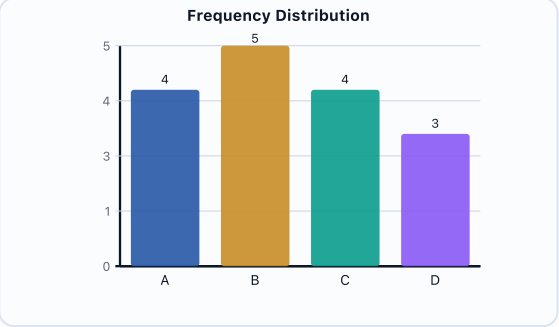
Mark Scheme & Examiner Solution (Histograms, Cumulative Frequency & Conditional Probability):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 11, d = 6$: $S_{15} = \frac{15}{2}[22 + 14 \times 6] = \frac{15}{2} \times 106 = 795$. **[M1][A1]**

- **Final Verified Answer:** 795.

5 MCQ

In Histograms, Cumulative Frequency & Conditional Probability, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 40x)\,\mathrm{d}x$.



A 152

B 168

C 65

D 128 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Histograms, Cumulative Frequency & Conditional Probability):

1. Integrate term by term: $\int (18x^2 + 40x)\,\mathrm{d}x = [6x^3 + 20x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 20(2)^2) - 0 = 48 + 80 = 128$. **[M1][A1]**

- **Final Verified Answer:** 128.

6 MCQ

In Histograms, Cumulative Frequency & Conditional Probability, consider the quadratic function $f_6(x) = x^2 - 22x + 211$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(18)$.

- **A** 229 ✓
- **B** 240
- **C** 139
- **D** 222

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Histograms, Cumulative Frequency & Conditional Probability):

1. Complete the square: $x^2 - 22x + 211 = (x-11)^2 - 121 + 211 = (x-11)^2 + 90$, giving minimum value $k = 90$. **[M1]**
2. Evaluate at $x = 18$: $f_6(18) = (7)^2 + 90 = 139$. **[M1]**
3. Sum the two quantities: $k + f_6(18) = 90 + 139 = 229$. **[A1]**

- **Final Verified Answer:** 229.

7 MCQ

The graph illustrates the cubic model $y_7(x) = 3x^3 - 11x^2 + 33x - 22$ studied in Histograms, Cumulative Frequency & Conditional Probability. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

- **A** 609
- **B** 433 ✓
- **C** 400
- **D** 83

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Histograms, Cumulative Frequency & Conditional Probability):

1. Differentiate term by term using the power rule: $\frac{\mathrm{d}y}{\mathrm{d}x} = 9x^2 - 22x + 33$. **[M1]**
2. Substitute $x = 8$: $\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=8} = 9(64) - 22(8) + 33 = 433$. **[M1][A1]**

- **Final Verified Answer:** 433.

8 MCQ

In Histograms, Cumulative Frequency & Conditional Probability, two displacement vectors are given by $\mathbf{u}_8 = \langle 11, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

- **A** 53
- **B** 170
- **C** 223 ✓
- **D** 293

WORKED SOLUTION & VERIFICATION

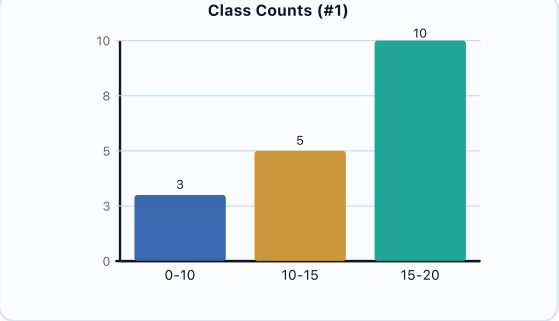
Mark Scheme & Examiner Solution (Histograms, Cumulative Frequency & Conditional Probability):

1. Compute the inner product: $\mathbf{u}_8 \cdot \mathbf{v}_8 = (11)(8) + (-7)(5) = 88 + (-35) = 53$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_8\|^2 = 11^2 + (-7)^2 = 170$. **[M1]**
3. Combine: $53 + 170 = 223$. **[A1]**

- **Final Verified Answer:** 223.

9 MCQ

In a grouped frequency table (#1), the class interval $10 < t \leq 15$ contains 5 observations. Calculate the frequency density plotted on a histogram for this class.



A 25

B 1.00

C 1 ✓

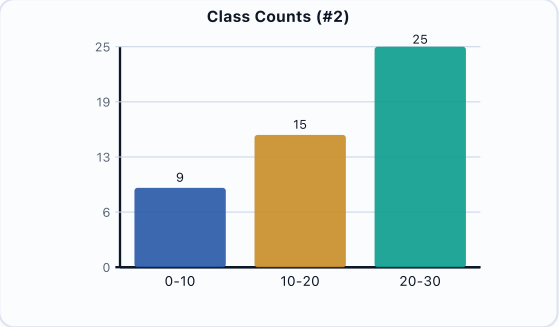
D 2.0

WORKED SOLUTION & VERIFICATION

Class width $= 15 - 10 = 5$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{5}{5} = 1$.

10 MCQ

In a grouped frequency table (#2), the class interval $10 < t \leq 20$ contains 15 observations. Calculate the frequency density plotted on a histogram for this class.



A 0.67

B 1.5 ✓

C 150

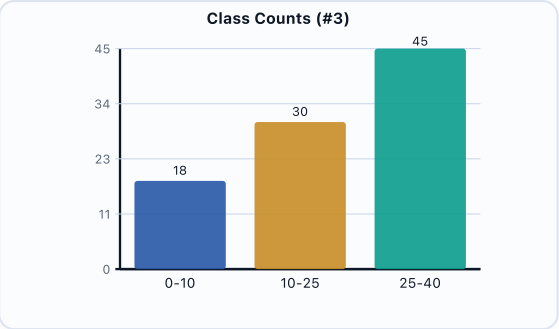
D 3.0

WORKED SOLUTION & VERIFICATION

Class width $= 20 - 10 = 10$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{15}{10} = 1.5$.

11 MCQ

In a grouped frequency table (#3), the class interval $10 < t \leq 25$ contains 30 observations. Calculate the frequency density plotted on a histogram for this class.



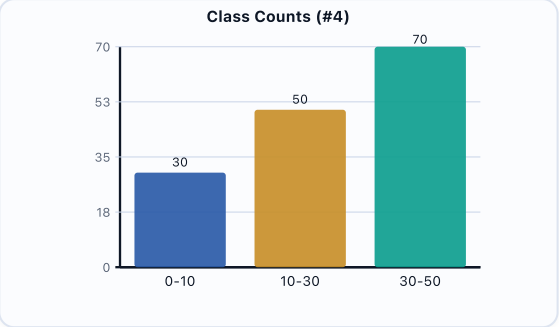
- **A** 2 ✓
- **B** 4.0
- **C** 450
- **D** 0.50

WORKED SOLUTION & VERIFICATION

Class width $= 25 - 10 = 15$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{30}{15} = 2$.

12 MCQ

In a grouped frequency table (#4), the class interval $10 < t \leq 30$ contains 50 observations. Calculate the frequency density plotted on a histogram for this class.



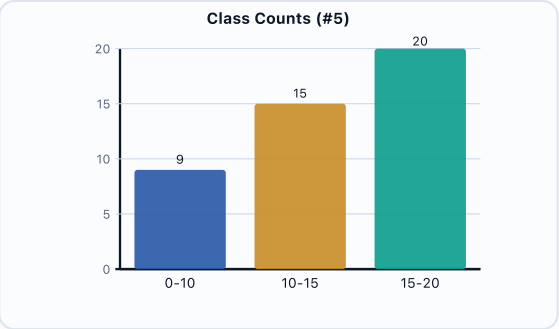
- **A** 5.0
- **B** 2.5 ✓
- **C** 0.40
- **D** 1000

WORKED SOLUTION & VERIFICATION

Class width $= 30 - 10 = 20$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{50}{20} = 2.5$.

13 MCQ

In a grouped frequency table (#5), the class interval $10 < t \leq 15$ contains 15 observations. Calculate the frequency density plotted on a histogram for this class.



A 3 ✓

B 6.0

C 0.33

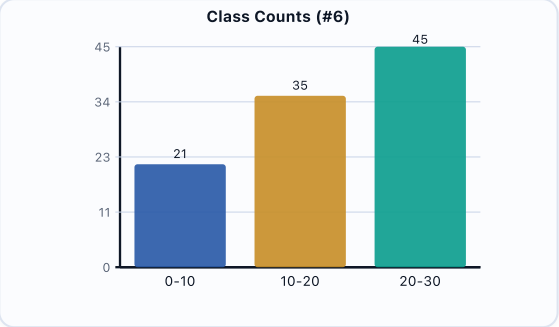
D 75

WORKED SOLUTION & VERIFICATION

Class width $= 15 - 10 = 5$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{15}{5} = 3$.

14 MCQ

In a grouped frequency table (#6), the class interval $10 < t \leq 20$ contains 35 observations. Calculate the frequency density plotted on a histogram for this class.



A 7.0

B 3.5 ✓

C 0.29

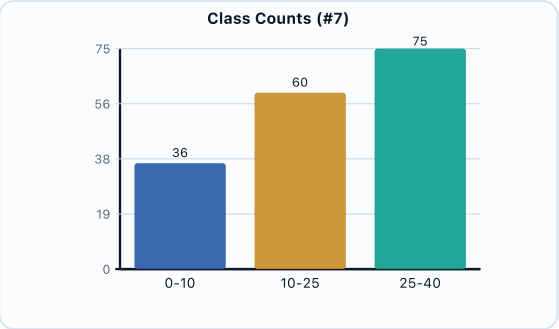
D 350

WORKED SOLUTION & VERIFICATION

Class width $= 20 - 10 = 10$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{35}{10} = 3.5$.

15 MCQ

In a grouped frequency table (#7), the class interval $10 < t \leq 25$ contains 60 observations. Calculate the frequency density plotted on a histogram for this class.



A 0.25

B 4 ✓

C 900

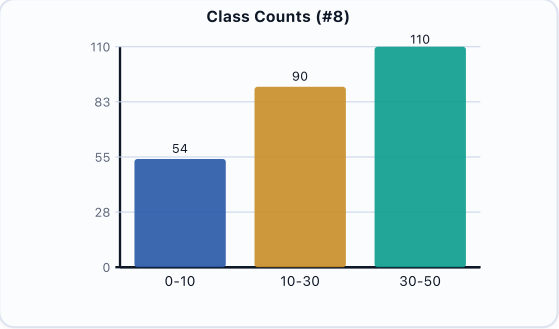
D 8.0

WORKED SOLUTION & VERIFICATION

Class width $= 25 - 10 = 15$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{60}{15} = 4$.

16 MCQ

In a grouped frequency table (#8), the class interval $10 < t \leq 30$ contains 90 observations. Calculate the frequency density plotted on a histogram for this class.



A 4.5 ✓

B 0.22

C 1800

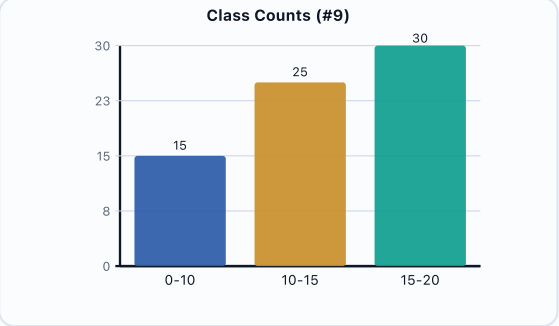
D 9.0

WORKED SOLUTION & VERIFICATION

Class width $= 30 - 10 = 20$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{90}{20} = 4.5$.

17 MCQ

In a grouped frequency table (#9), the class interval $10 < t \leq 15$ contains 25 observations. Calculate the frequency density plotted on a histogram for this class.



A 0.20

B 10.0

C 125

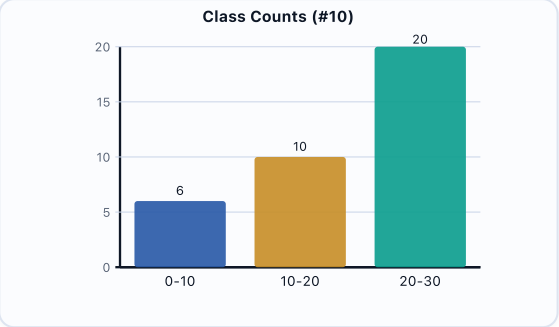
D 5 ✓

WORKED SOLUTION & VERIFICATION

Class width $= 15 - 10 = 5$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{25}{5} = 5$.

18 MCQ

In a grouped frequency table (#10), the class interval $10 < t \leq 20$ contains 10 observations. Calculate the frequency density plotted on a histogram for this class.



A 100

B 2.0

C 1.00

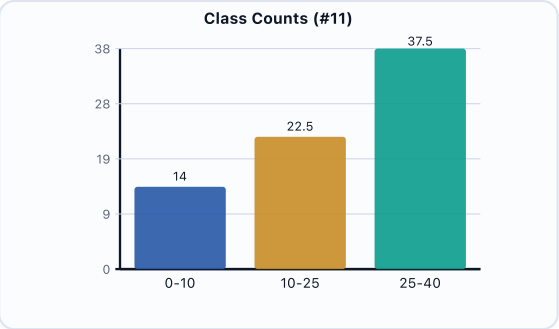
D 1 ✓

WORKED SOLUTION & VERIFICATION

Class width $= 20 - 10 = 10$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{10}{10} = 1$.

19 MCQ

In a grouped frequency table (#11), the class interval $10 < t \leq 25$ contains 22.5 observations. Calculate the frequency density plotted on a histogram for this class.



A 337.5

B 0.67

C 1.5 ✓

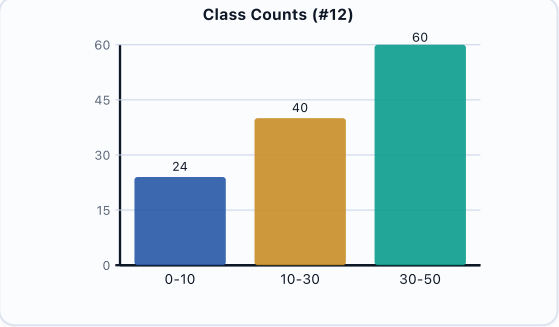
D 3.0

WORKED SOLUTION & VERIFICATION

Class width $= 25 - 10 = 15$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{22.5}{15} = 1.5$.

20 MCQ

In a grouped frequency table (#12), the class interval $10 < t \leq 30$ contains 40 observations. Calculate the frequency density plotted on a histogram for this class.



A 800

B 2 ✓

C 0.50

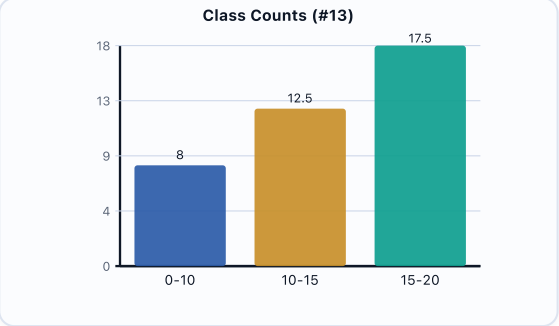
D 4.0

WORKED SOLUTION & VERIFICATION

Class width $= 30 - 10 = 20$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{40}{20} = 2$.

21 MCQ

In a grouped frequency table (#13), the class interval $10 < t \leq 15$ contains 12.5 observations. Calculate the frequency density plotted on a histogram for this class.



A 2.5 ✓

B 0.40

C 5.0

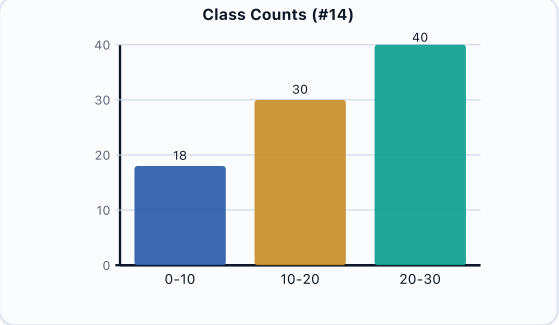
D 62.5

WORKED SOLUTION & VERIFICATION

Class width $= 15 - 10 = 5$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{12.5}{5} = 2.5$.

22 MCQ

In a grouped frequency table (#14), the class interval $10 < t \leq 20$ contains 30 observations. Calculate the frequency density plotted on a histogram for this class.



A 6.0

B 3 ✓

C 300

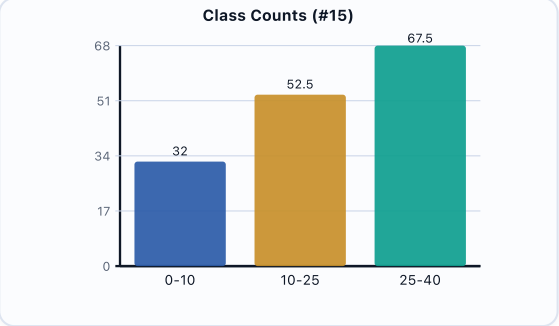
D 0.33

WORKED SOLUTION & VERIFICATION

Class width $= 20 - 10 = 10$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{30}{10} = 3$.

23 MCQ

In a grouped frequency table (#15), the class interval $10 < t \leq 25$ contains 52.5 observations. Calculate the frequency density plotted on a histogram for this class.



A 7.0

B 0.29

C 787.5

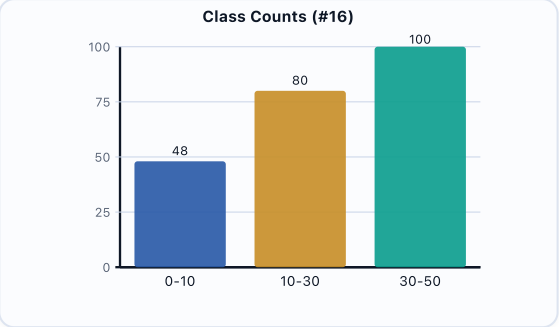
D 3.5 ✓

WORKED SOLUTION & VERIFICATION

Class width $= 25 - 10 = 15$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{52.5}{15} = 3.5$.

24 MCQ

In a grouped frequency table (#16), the class interval $10 < t \leq 30$ contains 80 observations. Calculate the frequency density plotted on a histogram for this class.



A 0.25

B 4 ✓

C 8.0

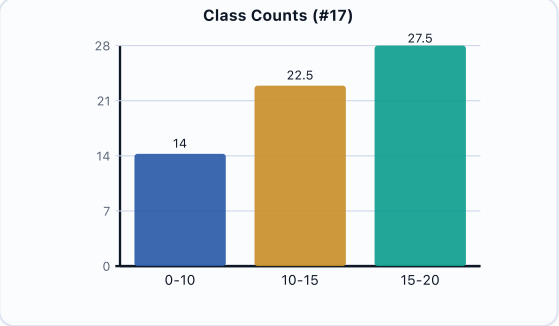
D 1600

WORKED SOLUTION & VERIFICATION

Class width $= 30 - 10 = 20$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{80}{20} = 4$.

25 MCQ

In a grouped frequency table (#17), the class interval $10 < t \leq 15$ contains 22.5 observations. Calculate the frequency density plotted on a histogram for this class.



A 112.5

B 9.0

C 4.5 ✓

D 0.22

WORKED SOLUTION & VERIFICATION

Class width $= 15 - 10 = 5$. Frequency density $= \frac{\text{Frequency}}{\text{Class Width}} = \frac{22.5}{5} = 4.5$.